

Research defense and next questions

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Presentation and challenge session

Without opening the old lesson, explain one key idea from Presentation and challenge session. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Independent reproduction check

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Writing the methods section

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Research defense and next questions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Research defense and next questions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Project candidate9/12, baseline10/12; reproduction matches. State result and a useful next question.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Study shows candidate12/16 versus baseline11/16; replay matches; no fresh population sample or user study. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: The next question follows evidence gaps; it must not be presented as an answer already obtained. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Project candidate9/12, baseline10/12; reproduction matches. State result and a useful next question.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final evidence pack: Project candidate9/12, baseline10/12; reproduction matches. State result and a useful next question. Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Research defense and next questions

Defend the next research question. A research defence identifies a observed weakness and a testable follow-up.

A frozen model is correct on 8/10 clear images and 4/10 shaded images. Source objects are grouped within each condition.

Defence: different condition slices

correct cases out of 10 per slice



Bar length shows the supplied value; the marked baseline is zero.

Research defense and next questions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. State slice outcomes

Clear is 80%; shaded is 40%.



2. Limit the interpretation

Shade sensitivity is plausible, but camera or object differences may also matter.



3. Design follow-up

Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

CHANGE ONE THING

Change cameras as well as lighting in the follow-up.

Explain the changed result using the matching lesson.

Compute the clear and shaded rates and identify alternative explanations. How does the same-object, one-camera follow-up help, and why does changing cameras again reintroduce confounding?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend the next research question. A research defence identifies a observed weakness and a testable follow-up.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Photograph the same held-out objects under both light conditions using one camera and fixed scoring.

1. Design follow-up
2. Limit the interpretation
3. State slice outcomes

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Design follow-up Photograph the same held-out objects under both light conditions using one camera and fixed scoring. A defence should explain results, uncertainty and what would discriminate between explanations next. A controlled paired follow-up is more informative than simply collecting extra easy images.

Practice 2: explain the deciding evidence

Changed-case practice: Change cameras as well as lighting in the follow-up. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The design again entangles two conditions.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The design again entangles two conditions. A camera difference provides another explanation for any performance gap.

Repair a missing building block

Review Presentation and challenge session (ai-7/35). Goal: Explain the mechanism and terminology of presentation and challenge session. Short practice: Answer the camera-B challenge using only the twelve camera-A records. Why is one displayed camera-B success a demonstration rather than broad evidence of camera-B reliability?. Return to this week and explain what you changed.

Review Independent reproduction check (ai-7/33). Goal: Explain the mechanism and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the mechanism and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow. Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

